17/10/25 **Bivariate normal distribution**

$X_1 \sim N(\mu_1, \sigma_1^2)$ $X_2 \sim N(\mu_2, \sigma_2^2)$ indep

$X_1 + X_2 \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$

$Y = aX_1 + bX_2$

$M_Y(t) = M_{X_1}(at) \cdot M_{X_2}(bt) = e^{a\mu_1 t + \frac{a^2\sigma_1^2 t^2}{2}} \cdot e^{b\mu_2 t + \frac{b^2\sigma_2^2 t^2}{2}}$

$= e^{(a\mu_1 + b\mu_2)t + \frac{(a^2\sigma_1^2 + b^2\sigma_2^2)t^2}{2}}$

$Y \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$

$U = a_1X_1 + a_2X_2$

$V = b_1X_1 + b_2X_2$

$U \sim$ normal $N(a_1\mu_1 + a_2\mu_2, a_1^2\sigma_1^2 + a_2^2\sigma_2^2)$

$V \sim$ normal $N(b_1\mu_1 + b_2\mu_2, b_1^2\sigma_1^2 + b_2^2\sigma_2^2)$

$aU + bV \sim$ normal $N(\,,\,)$

Definition: U, V are said to be bivariate normal if every linear combination of U and V is normal.

→ (U, aU) is bivariate normal — "Trivial"

U: normal.

General form for joint PDF $X_1, X_2 \sim N(0,1)$ indep.

$f_{X_1X_2}(x_1, x_2) = \frac{1}{2\pi} e^{\frac{-(x_1^2+x_2^2)}{2}}$

$U = aX_1 + bX_2 + c$

$V = dX_1 + eX_2 + f$

$(ae - bd \neq 0)$

$$f_{UV}(u,v) = ?$$

Simpler (with LOA)

$$U = aX_1 + c \qquad x_1 = \frac{u-c}{a}$$

$$V = dX_1 + eX_2 + f \qquad x_2 = \left(\frac{v-f}{e}\right) - \frac{d}{e}\left(\frac{u-c}{a}\right)$$

$\mu_U = c$

$\sigma_U = |a|$

$\mu_V = f$

$\sigma_V^2 = d^2 + e^2$

$$\rho_{UV} = \frac{ad}{a\cdot\sqrt{d^2+e^2}}$$

$$f_{UV}(u,v) = \frac{1}{2\pi} e^{\frac{-\left(\left(\frac{u-c}{a}\right)^2 + \left(\frac{v-f}{e}\right)^2 + \frac{d^2}{e^2}\left(\frac{u-c}{a}\right)^2 - \frac{2d}{e}\left(\frac{v-f}{e}\right)\left(\frac{u-c}{a}\right)\right)}{2}}$$

$$= \frac{1}{2\pi|ae|} e^{\frac{-\left(\left(1+\frac{d^2}{e^2}\right)\left(\frac{u-c}{a}\right)^2 - 2\frac{d}{e}\left(\frac{u-c}{a}\right)\left(\frac{v-f}{e}\right) + \left(\frac{v-f}{e}\right)^2\right)}{2}}$$

$$= \frac{1}{2\pi|ae|} e^{-\left[\left(\frac{u-c}{a}\right)^2 - 2\frac{d}{\sqrt{d^2+e^2}}\left(\frac{u-c}{a}\right)\left(\frac{v-f}{\sqrt{d^2+e^2}}\right) + \left(\frac{v-f}{\sqrt{d^2+e^2}}\right)^2\right] / 2\cdot e^2/(e^2+d^2)}$$

$$= \frac{1}{2\pi|ae|} e^{-\left[\left(\frac{u-\mu_U}{\sigma_U}\right)^2 - 2\rho_{UV}\left(\frac{u-\mu_U}{\sigma_U}\right)\cdot\left(\frac{v-\mu_V}{\sigma_V}\right) + \left(\frac{v-\mu_V}{\sigma_V}\right)^2\right] / 2\cdot(1-\rho_{UV}^2)}$$

$$f_{UV}(u,v) = \frac{1}{2\pi\sigma_U\sigma_V\sqrt{1-\rho_{UV}^2}}$$

"Bivariate normal"

"Standard" $\mu_U = \mu_V = 0 \quad \sigma_U = \sigma_V = 1 \quad \rho_{UV} = \rho$

$|\rho| < 1$

$$f_{UV}(u,v) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{\frac{-\left(u^2 - 2\rho uv + v^2\right)}{2\sqrt{1-\rho^2}}}$$

$\rho = 0$ uncorrelated $\Rightarrow$ independent



